

Roll No.

ODD END SEMESTER EXAMINATION DECEMBER 2024

Name of the Course: BTech

Semester: 7

Name of the Paper: Natural Language Processing with Big Data

Paper code: TCS 771

Time: 3 Hours

Maximum Marks: 100

Note:-

- i. All questions are compulsory.
- ii. Answer any two sub questions among a, b & c in each main question.
- iii. Total marks in each main question are twenty.
- iv. Each question carries 10 marks.

Q1.

- a) Describe four techniques for converting text data into features in natural language processing. [CO1]
- b) Explain the vector space model and cosine similarity. How are they applied in text analysis? [CO1]
- c) With an example, explain how an n-gram model can improve the performance of an NLP task. [CO2]

Q2

- a) Write Python code to calculate TF and TF-IDF weights for three sentences stored as strings in variables s1,s2,s1,s2,and s3s3. [CO2]
- b) Analyze two linear classifiers commonly used in NLP. How do the `fit()` and `predict()` methods differ? [CO3]
- c) Define pretrained language models and explain model fine-tuning with examples. [CO5]

Q3.

- a) Differentiate between contextual embeddings and static embeddings. Provide an example of each type. [CO3]

b) Explain the concepts of language modeling and next sentence prediction in the BERT model. [CO5]

c) How are recurrent neural networks (RNNs) used as language models? Illustrate with an example. [CO4]

Q4.

a) Describe the Naïve Bayes classifier's application in spam detection. Why is Laplace smoothing required? [CO2]

b) Given 100 samples:

- Positive samples: 60 (50 classified correctly),
- Negative samples: 40 (35 classified correctly),
- Construct the confusion matrix and calculate precision, recall, and F1-score. [CO2]

c) Explain the concepts of cross-validation and stratified sampling with suitable examples. [CO2]

Q5.

a) Discuss four applications of deep learning in natural language processing. [CO6]

b) Define and explain the importance of loss functions, gradients, and backpropagation in neural network training. [CO3]

c) Describe the self-attention mechanism and its role in natural language processing with an example. [CO5]